

A FAST LEAST SQUARES ALGORITHM FOR SEPARATING TRILINEAR MIXTURES

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ABSTRACT

In this paper an algorithm called COMFAC (COMplex parallel FACTor analysis) is developed for fitting the trilinear parallel factor analysis model to data arising from e.g., DS-CDMA signals or joint azimuth-elevation estimation. The algorithm is based on combining and improving a number of auxiliary sub-routines with the purpose of providing the shortest possible computation time. The different steps in the overall COMFAC algorithm are described and the algorithm is applied to different relevant problems.

1. INTRODUCTION

Three-way factor analysis and the trilinear PARAFAC model in particular often underlies the problem of blind separation of DS-CDMA signals in unknown multipath using an antenna array [17], joint azimuth-elevation estimation [5], joint angle-delay estimation, and the subspace fitting formulation of multiple-parameter/multiple-invariance ESPRIT [5]. In communications and signal processing the data and model parameters are often complex-valued, due to in-phase/quadrature processing at the transmitter/receiver. As a result, there is a need to develop efficient algorithms for fitting the complex PARAFAC model. Furthermore most available PARAFAC algorithms are either very slow in certain circumstances or very complicated to use. In this paper a new algorithm is developed with the purpose of circumventing these problems. The algorithm is thus not only an extension to the complex case but also provides a substantially improved algorithm for the real-valued problem.

Historically the approach of Alternating Least Squares regression (ALS) has been the common cornerstone of most PARAFAC algorithms [1,3,8]. Alternating least squares regression is attractive because it is simple to apply, it guarantees convergence and it is easily modified so that restrictions can be imposed on the parameters [4]. The main drawback, however, is that convergence is often linear and the algorithm is quite sensitive to local minima and possibly intermediate collinearity problems: so-called swamps. The algorithm presented here circumvents these problems by using a very fast and precise initialization.

In our algorithm, COMFAC, the data to be modeled are initially compressed in an optimal sense [3] in order to speed up the subsequent iterative fitting. The so-called CANDELINC (CANonical DEcomposition with LINEar Constraints) procedure [7] combined with a fast algorithm for fitting the Tucker3 model [1,11] is used for this purpose. From the compressed data an approximate PARAFAC model is fitted using the Generalized Rank Annihilation Method (GRAM). This approximate model is used as input to the core algorithm, namely the iterative

PARAFAC algorithm. The output is the set of model parameters of the fitted compressed data. These are converted to the original uncompressed domain by means of a simple matrix multiplication. The last step of the COMFAC algorithm is to use this solution as input to a PARAFAC algorithm working in the original non-compressed space. This serves as a guard against artifacts introduced by fitting in the compressed space.

In the following we first describe the generic problem to be solved. Then the details of the COMFAC algorithm are given. Finally some example applications are given. A MATLAB implementation is available at <http://newton.foodsci.kvl.dk>.

2. DEFINING THE PROBLEM

An $I \times J \times K$ three-way array $\underline{\mathbf{X}}$ is given with typical element x_{ijk} , $i = 1, \dots, I, j = 1, \dots, J, k = 1, \dots, K$ and $x_{ijk} \in \mathbb{C}$. Such an array may be graphically depicted as in Figure 1, where the three different directions are called the first, second and third mode respectively.

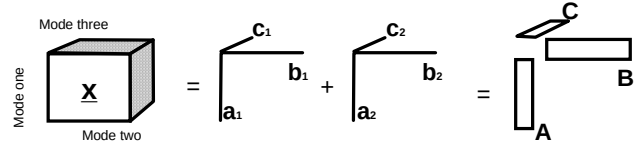


Figure 1. A three-way array shown to the left and two different representations of a two-component PARAFAC model shown to the right.

We seek the PARAFAC model

$$x_{ijk} = \sum_{f=1}^F a_{if} b_{jf} c_{kf} + e_{ijk} \quad (1)$$

where a_{if} , b_{jf} and c_{kf} are the parameters to estimate, e_{ijk} are the residuals and F is the number of components sought. Let $\mathbf{X}_k$ be the k th horizontal slab of the three-way array, thus the $I \times J$ matrix obtained by fixing the third mode index at k . Then the model may equivalently be written

$$\mathbf{X}_k = \mathbf{A} \mathbf{D}_k \mathbf{B}^T + \mathbf{E}_k \quad (2)$$

where $\mathbf{A}$ is an $I \times F$ matrix with elements a_{if} , $\mathbf{B}$ is a $J \times F$ matrix with elements b_{jf} and $\mathbf{D}_k$ is a diagonal matrix holding the k th row of the $K \times F$ matrix $\mathbf{C}$ with elements c_{kf} in its diagonal. In order to fit this model in a least squares sense we minimize the corresponding loss function

$$\sigma = \sum_{k=1}^K \left\| \mathbf{X}_k - \mathbf{A} \mathbf{D}_k \mathbf{B}^T \right\|_F^2 \quad (3)$$

where $\|\bullet\|_F^2$ is the Frobenius norm. In the following the names $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ will be generally used for describing (oblique) loading matrices in the first, second and third mode respectively, whereas the matrices $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{Z}$ will designate unitary bases for subspaces of these modes.

3. COMFAC ALGORITHM

3.1 Step one: Compression

The first step in the COMFAC algorithm is to compress the array $\mathbf{X}$ in order to save computation time. For this purpose a lossy but effective compression method is used which utilizes a theorem of [7] that directly guarantees optimal results if good truncated bases for the respective modes are available. The compression method was originally described in [3] for the case of real-valued arrays.

Assume that a basis for the systematic variation in the first mode is given, i.e. a basis that spans the variation of an ideal array without noise or model mis-specification. Let $\mathbf{U}$ ($I \times L$) be a unitary matrix representing this basis. Since the first mode loadings, $\mathbf{A}$, of the PARAFAC model spans the systematic variation it must hold that

$$\mathbf{A} = \mathbf{U} \mathbf{R} \quad (4)$$

where $\mathbf{R}$ is an $L \times F$ matrix. Similarly define a unitary matrix $\mathbf{V}$ ($J \times M$) as a basis for the systematic variation in the second mode and a matrix $\mathbf{Z}$ ($K \times N$) as a basis for the systematic variation in the third mode. It then holds that

$$\mathbf{B} = \mathbf{V} \mathbf{P} \text{ and } \mathbf{C} = \mathbf{Z} \mathbf{Q} \quad (5)$$

where $\mathbf{P}$ is an $M \times F$ and $\mathbf{Q}$ an $N \times F$ matrix. The CANDELINC theorem states the following. If the PARAFAC model of Equation 3 is sought under the linear constraints in Equations 4-5 then one need only estimate the parameter matrices $\mathbf{R}$, $\mathbf{P}$ and $\mathbf{Q}$. Furthermore, estimating these parameters amounts to fitting a PARAFAC model to the $L \times M \times N$ array $\mathbf{G}$ found as

$$\mathbf{G} = \mathbf{U}^T \mathbf{X} (\mathbf{Z} \otimes \mathbf{V}) \quad (6)$$

where $\mathbf{G}$ is the array $\mathbf{G}$ unfolded to an $L \times MN$ two-way matrix by concatenating the N slabs of size $L \times M$ next to each other. Likewise $\mathbf{X}$ is the $I \times J \times K$ array $\mathbf{X}$ unfolded to an $I \times JK$ matrix. The operator $\otimes$ is the Kronecker product. The compressed array $\mathbf{G}$ represents the regression of $\mathbf{X}$ onto $\mathbf{U}$ in the first mode, onto $\mathbf{V}$ in the second mode and onto $\mathbf{Z}$ in the third mode. The array may therefore be interpreted as the coordinates of $\mathbf{X}$ in the truncated and rotated system defined by the three bases. If $\mathbf{X}$ is completely within the spaces defined by these three bases the transformation is loss-less and the variance in $\mathbf{X}$ will be preserved in $\mathbf{G}$. The array $\mathbf{G}$ is much smaller than the original array $\mathbf{X}$ especially if some modes in $\mathbf{X}$ are very high dimensional. Hence fitting a PARAFAC model to $\mathbf{G}$ is computationally much simpler than fitting a model to $\mathbf{X}$.

In order for the compression to be useful it is necessary to be able to find suitable bases in a simple fashion. Consider the definition of the compressed array $\mathbf{G}$ in Equation 6. The array $\mathbf{G}$ is the part of $\mathbf{X}$ within the subspaces defined by the three bases. The projection of $\mathbf{X}$ onto these bases is given

$$\mathbf{U} \mathbf{U}^T \mathbf{X} (\mathbf{Z} \otimes \mathbf{V}) \mathbf{V}^T = \mathbf{U} \mathbf{U}^T \mathbf{X} (\mathbf{Z} \otimes \mathbf{V}) (\mathbf{Z} \otimes \mathbf{V})^T = \mathbf{U} \mathbf{G} (\mathbf{Z} \otimes \mathbf{V})^T \quad (7)$$

Given $\mathbf{X}$ and the definition of $\mathbf{G}$ the optimal bases in a least squares sense must be those that minimize

$$\left\| \mathbf{X} - \mathbf{U} \mathbf{G} (\mathbf{Z} \otimes \mathbf{V})^T \right\|_F^2. \quad (8)$$

subject to $\mathbf{U}$, $\mathbf{V}$ and $\mathbf{Z}$ all being unitary. This is seen to be the three-mode principal component analysis model also called the Tucker3 model [1,11]. The compressed array $\mathbf{G}$ is called the core array in the terminology of multi-way analysis and the Tucker3 model thus directly provides the optimal bases as well as the compressed array. The solution to fitting the Tucker3 model has been devised in [10] and a fast implementation was provided in [1]. This Tucker3 algorithm can easily be extended to the complex-valued case as outlined below:

Algorithm TUCKER3-ALS

Initialize $\mathbf{V}$ and $\mathbf{Z}$

1. Let $\mathbf{U}$ equal the first L left singular vectors of a singular value decomposition of the matrix $\mathbf{X}^{(I \times JK)} (\mathbf{Z} \otimes \mathbf{V})$
2. Let $\mathbf{V}$ equal the first M left singular vectors of a singular value decomposition of the matrix $\mathbf{X}^{(J \times IK)} (\mathbf{Z} \otimes \mathbf{U})$
3. Let $\mathbf{Z}$ equal the first N left singular vectors of a singular value decomposition of the matrix $\mathbf{X}^{(K \times IJ)} (\mathbf{V} \otimes \mathbf{U})$

Go to step 1 until relative change in fit is small and calculate the core as $\mathbf{G} = \mathbf{U}^T \mathbf{X} (\mathbf{Z} \otimes \mathbf{V})$

Further algorithmic details are omitted for brevity but may be found in [1]. Two important aspects of the practical use of the Tucker3 model are the choice of number of components and the convergence criterion to use. These matters are greatly interdependent and the optimal settings will vary depending on the size of data as well as on the signal-to-noise ratio and the appropriateness of the subsequent PARAFAC model. For compression of the data used in this paper it has been found useful to use only four iterations in the Tucker3 algorithm instead of iterating to a fixed convergence criterion. The number of components in each mode is set to $F + 2$ where F is the number of components in the PARAFAC model. If a mode has dimension less than $F + 2$, the number of components in that mode is reduced correspondingly.

3.2 Step two: Initialization

After compression, a PARAFAC model is fitted to the array $\mathbf{G}$. The PARAFAC algorithm is iterative and good starting values are beneficial in keeping computation time low. For noise-free data with no model error an exact solution can be calculated directly by posing the fitting problem as a generalized eigenvalue

problem much in line with ESPRIT [15]. In case of fallible data the solution will not be the least squares solution but for well-posed problems it will be close. The direct solution is known in chemometrics as the Generalized Rank Annihilation Method or GRAM. The basis of GRAM has been described in [9,13]. The method is more or less identical to ESPRIT and was later further developed in a chemical context in e.g. [16].

The following theorem establishes the GRAM method as well as proof of uniqueness of the PARAFAC model. The proof provides sufficient conditions though it is known that uniqueness of the PARAFAC model exists also in cases not covered by GRAM and ESPRIT [12]. Although the COMFAC initialization involves fitting the transformed array $\underline{\mathbf{G}}$, GRAM will be described using the notation that the data are held in $\underline{\mathbf{X}}$ and that the parameter matrices are named according to the PARAFAC model in the original domain. This is done to stress that GRAM is a general method, which may also be used in the original domain if so wished.

Theorem 1. Generalized Rank Annihilation Method

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be matrices of size $I \times J$ with elements $x_{ijk} \in \mathbb{C}$. Let the matrices be parameterized as

$$\mathbf{X}_1 = \mathbf{A}\mathbf{D}_1\mathbf{B}^T \text{ and } \mathbf{X}_2 = \mathbf{A}\mathbf{D}_2\mathbf{B}^T \quad (9)$$

where $\mathbf{A}$ ($I \times F$) and $\mathbf{B}$ ($J \times F$) are possibly complex matrices and $\mathbf{D}_1$ and $\mathbf{D}_2$ are possibly complex diagonal matrices. Thus, this is a PARAFAC model according to Eq. 2 with a third mode of dimension two. Assume that $\mathbf{A}$ and $\mathbf{B}$ are of full column rank, and that the diagonal matrix $\mathbf{D}_1^{-1}\mathbf{D}_2$ is distinct, i.e., have no identical or zero diagonal elements. Let the $I \times F$ matrix $\mathbf{U}$ be a unitary matrix spanning the row-space of $\mathbf{X}_1$ and $\mathbf{X}_2$ and the $J \times F$ unitary matrix $\mathbf{V}$ span the column-space. Define

$$\mathbf{S}_1 = \mathbf{U}^T \mathbf{X}_1 \mathbf{V} \text{ and } \mathbf{S}_2 = \mathbf{U}^T \mathbf{X}_2 \mathbf{V} \quad (10)$$

hence $\mathbf{S}_1$ and $\mathbf{S}_2$ are both $F \times F$ matrices. Let

$$\mathbf{S}_2 \mathbf{E} = \mathbf{S}_1 \mathbf{E} \mathbf{T} \quad (11)$$

be the generalized eigen-value problem for $\mathbf{S}_1$ and $\mathbf{S}_1$ where $\mathbf{E}$ is the generalized eigenvector matrix and $\mathbf{T}$ the diagonal matrix holding the eigenvalues. Then the parameters of the PARAFAC model may be determined as

$$\mathbf{A} = \mathbf{U}\mathbf{T}, \mathbf{B} = \mathbf{V}\mathbf{\Delta}, \mathbf{D}_1 = \mathbf{\Gamma}^{-1}\mathbf{S}_1\mathbf{\Delta}^{-1} \text{ and } \mathbf{D}_2 = \mathbf{\Gamma}^{-1}\mathbf{S}_2\mathbf{\Delta}^{-1}, \quad (12)$$

where

$$\mathbf{\Gamma} = \mathbf{S}_1 \mathbf{E} \text{ and } \mathbf{\Delta} = \mathbf{E}^{-1}. \quad (13)$$

The proof of the above theorem is left out for brevity but follows closely the proof of ESPRIT, GRAM and similar methods. ■

The theorem applies to the case of noise-free data but an algorithm for approximating the solution in case of data that does not conform exactly to the model is easily implemented as below.

Algorithm Tucker2 - GRAM

Let $\mathbf{U}$ be an $I \times F$ unitary matrix providing the best truncated basis with respect to $\mathbf{X}_1$ as well as $\mathbf{X}_2$. Let the $J \times F$ matrix, $\mathbf{V}$ be a truncated basis for the column-space of both data arrays. In the

noise-free case these basis matrices may be found from any of the arrays but for noisy data this is not the case. The most reasonable way to find the basis matrices is to determine these such that they simultaneously provide maximal spanning subspaces for both data sets. This may be formulated

$$\|\mathbf{U}\mathbf{U}^+ \mathbf{X}_1 (\mathbf{V}\mathbf{V}^+)^T\|_F^2 + \|\mathbf{U}\mathbf{U}^+ \mathbf{X}_2 (\mathbf{V}\mathbf{V}^+)^T\|_F^2 =$$

$$\|\mathbf{U}\mathbf{U}^+ [\mathbf{X}_1 \mathbf{X}_2] (\mathbf{Z}\mathbf{Z}^+ \otimes \mathbf{V}\mathbf{V}^+)^T\|_F^2 =$$

$$\|\mathbf{U}\mathbf{U}^+ [\mathbf{X}_1 \mathbf{X}_2] (\mathbf{Z}^+ \otimes \mathbf{V}^+) (\mathbf{Z} \otimes \mathbf{V})^T\|_F^2 =$$

$$\|\mathbf{U} [\mathbf{S}_1 \mathbf{S}_2] (\mathbf{Z} \otimes \mathbf{V})^T\|_F^2$$

where $\mathbf{Z}$ is a 2 by F matrix of ones and $\mathbf{S}_k = \mathbf{U}^+ \mathbf{X}_k \mathbf{V}$. The unitary matrices $\mathbf{U}$ and $\mathbf{V}$ maximizing this problem will asymptotically provide the best basis vectors. These can be found by means of the so-called Tucker2 model, which is equivalent to a Tucker3 model with one mode not being reduced. Given these truncated bases the parameters of the model may be calculated as described above.

Several papers have described GRAM-like methods and compared them with a least squares PARAFAC algorithm for their use in parameter estimation in chemistry. Generalized rank annihilation is mostly found to be inferior to the least squares PARAFAC model but if the data have a relatively high signal-to-noise ratio and the structural model is appropriate and correctly specified the two models are often close. The extension to more than two slabs is not straightforward. Several different approaches have been proposed in the literature for sub-optimal extensions. For brevity the extension suggested here will not be described in detail. The extension is based on defining two pseudo-slabs from the original K slabs and then subject these two pseudo-slabs to GRAM. This provides the parameters in two of the modes and the parameters in the last mode can subsequently be found by a single least squares regression step. The two slabs are found such that i) they are both in the same F -dimensional subspace and thus optimally amenable to a subsequent posing of the two-slab eigenvalue problem, and ii) they provide the two pseudo-slabs that are most descriptive of all slabs in a least squares sense. The slabs may be found by fitting a 2 by F by F Tucker3 model and thus unlike for example TLS-ESPRIT the subspace as well as the two pseudo-samples are computed simultaneously.

3.3 Step three: Fitting

In this paragraph an algorithm will be described for fitting the PARAFAC model in the compressed space. It is based on a Gauss-Newton approach and related to the PARAFAC algorithm described e.g., in [14]. The algorithm described here is strictly monotonically convergent though, and does not need any additional parameter fine-tuning. The algorithm is very fast provided that the initial solution is close to the true solution. This is ensured by the initialization described in the former paragraph. A distinguishing feature is that it is relatively memory

demanding, but working in the compressed domain ensures that the size of working arrays is kept small. As for GRAM, though, we will describe the algorithm in terms of the problem in the original domain.

Consider the PARAFAC model posed as

$$\mathbf{X} = \mathbf{A}(\mathbf{C}|\otimes|\mathbf{B})^T + \mathbf{E} \quad (14)$$

where the product

$$\mathbf{C}|\otimes|\mathbf{B} = [\mathbf{c}_1 \otimes \mathbf{b}_1 \ \mathbf{c}_2 \otimes \mathbf{b}_2 \ \dots \ \mathbf{c}_F \otimes \mathbf{b}_F] \quad (15)$$

is so-called Khatri-Rao product and $\mathbf{c}_f \otimes \mathbf{b}_f$ is the Kronecker product of the f th column of $\mathbf{B}$ and $\mathbf{C}$. The matrix $\mathbf{X}$ is the three-way array unfolded to a matrix as described earlier. Let $\mathbf{A}_r$, $\mathbf{B}_r$ and $\mathbf{C}_r$ be any given interim solution. It follows that

$$\begin{aligned} \mathbf{X} &= \mathbf{A}(\mathbf{C}|\otimes|\mathbf{B})^T + \mathbf{E} = \\ &(\mathbf{A}_r + \Delta\mathbf{A})((\mathbf{C}_r + \Delta\mathbf{C})|\otimes|(\mathbf{B}_r + \Delta\mathbf{B}))^T + \mathbf{E}. \end{aligned} \quad (16)$$

Assume that the interim solution is close to the true solution, hence $\Delta\mathbf{A}$, $\Delta\mathbf{B}$ and $\Delta\mathbf{C}$ are small. We may therefore assume that all higher-order differences such as $\Delta a_{if}\Delta b_{jf}$ are negligible. It then follows that

$$(\mathbf{A}_r + \Delta\mathbf{A})((\mathbf{C}_r + \Delta\mathbf{C})|\otimes|(\mathbf{B}_r + \Delta\mathbf{B}))^T \approx$$

$$\mathbf{A}_r(\mathbf{C}_r|\otimes|\mathbf{B}_r)^T + \Delta\mathbf{A}(\mathbf{C}_r|\otimes|\mathbf{B}_r)^T +$$

$$\mathbf{A}_r(\mathbf{C}_r|\otimes|\Delta\mathbf{B})^T + \mathbf{A}_r(\Delta\mathbf{C}|\otimes|\mathbf{B}_r)^T.$$

Setting

$$\mathbf{Y} = \mathbf{X} - \mathbf{A}_r(\mathbf{C}_r|\otimes|\mathbf{B}_r)^T \quad (17)$$

it follows that

$$\mathbf{Y} = \Delta\mathbf{A}(\mathbf{C}_r|\otimes|\mathbf{B}_r)^T + \mathbf{A}_r(\mathbf{C}_r|\otimes|\Delta\mathbf{B})^T + \mathbf{A}_r(\Delta\mathbf{C}|\otimes|\mathbf{B}_r)^T + \mathbf{E} \quad (18)$$

which may be formulated

$$\text{vec}\mathbf{Y} = \mathbf{H}\Delta\mathbf{f} + \mathbf{E} \quad (19)$$

where $\Delta\mathbf{f}$ is given

$$\Delta\mathbf{f} = \begin{bmatrix} \text{vec}\Delta\mathbf{A} \\ \text{vec}\Delta\mathbf{B} \\ \text{vec}\Delta\mathbf{C} \end{bmatrix} \quad (20)$$

and the $IJK \times (I+J+K)F$ matrix $\mathbf{H}$ is sparse and defined in the following way

$$h_{(j-1)I+(k-1)IJ+i,(f-1)I+i} = b_{j,f}c_{k,f}$$

$$h_{(j-1)I+(k-1)IJ+i,iF+(f-1)I+i} = a_{i,f}c_{k,f}$$

$$h_{(j-1)I+(k-1)IJ,(I+J)F+(f-1)I+i} = a_{i,f}b_{j,f}$$

where the subscript r has been skipped for brevity. Assume that $\mathbf{H}$ has full column rank, i.e., $(I+J+K)F \geq IJK$. This will be the case for most problems. If the assumption is not met, the algorithm may be replaced with the alternating least squares algorithm described next. The least squares solution to the problem in Eq. 19 is simply

$$\Delta\mathbf{f} = \mathbf{H}^+ \text{vec}\mathbf{Y} \quad (21)$$

and if the higher-order differences are small $\Delta\mathbf{f}$ it will provide a good update of the interim loading vectors. The basic idea in the Gauss-Newton algorithm is simply to repeat this updating of the loading matrices until the relative change in fit is small. It may happen, though, that an update causes the fit to decrease possibly leading to overall divergence of the algorithm. In order to circumvent this it is customary to do a line-search in the direction of $\Delta\mathbf{f}$ to find a point that increases the fit. Monitoring the fit as a function of the length-parameter in the line-search it is, however frequently observed that the fit in fact decreases for every positive value of the length-parameter. The cause is likely to be that the higher-order differences are not negligible and the direction defined by $\Delta\mathbf{f}$ therefore misleading. Instead the following approach is suggested if an interim update decreases the fit. The interim solution is skipped and instead the PARAFAC-ALS algorithm described next is used for a few iterations. Since the convergence of the PARAFAC-ALS algorithm is strictly monotonic, the subsequent update of the loading matrices *will* increase the fit and thus the overall algorithm will converge. The algorithm PARAFAC-GN is outlined below

Algorithm PARAFAC-GN

Initialize $\mathbf{A}_0$, $\mathbf{B}_0$ and $\mathbf{C}_0$ using GRAM

$r = 0$;

$n_{it} = 1$;

Set σ_r according to Eq. 3 using the current parameters

0 Set $r = r + 1$ and define the $IJK \times (I+J+K)F$ sparse matrix $\mathbf{H}$

1 $\Delta\mathbf{f} = \mathbf{H}^+ \text{vec}\mathbf{Y}$

2 Update $\mathbf{A}_r$, $\mathbf{B}_r$ and $\mathbf{C}_r$

3 Calculate σ_r

If $\sigma_r > \sigma_{r-1}$ do n_{it} PARAFAC-ALS iterations, set $n_{it} = 2n_{it}$

Otherwise if the relative change in σ_r is small stop.

Otherwise go to step one.

3.4 Step four: Decompressing

After fitting the F -component PARAFAC model to the compressed array using the PARAFAC-GN algorithm, the loading matrices in the compressed spaces are obtained. These are transformed into the spaces of the original array by means of the transformations in Equation 4-5 and a solution to the posed PARAFAC problem is obtained. To verify that this is the least squares solution a PARAFAC model is fitted to the raw data using the current PARAFAC parameters as initial values. Since computations are relatively expensive in the domain of the raw data and since only very few iterations are necessary to obtain convergence it is sensible to use an algorithm with a relatively low complexity per iteration. The well-established PARAFAC-ALS algorithm provides such an algorithm. It is based on alternately updating $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ conditionally on the remaining parameters. For each loading matrix the sub-problem to solve is a linear least squares regression problem and such an ALS-algorithm for fitting PARAFAC has been devised, e.g., in [1,3,81]. The alternating least squares algorithm for complex data follows directly as

Algorithm PARAFAC-ALS

Initialize $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ using PARAFAC-GN

$r = 0$;

$n_{it} = 1$;

Set σ_r according to Eq. 3 using the current parameters

0 Set $r = r + 1$

$$1 \quad \mathbf{A} = \sum_{k=1}^K \mathbf{X}_k \overline{\mathbf{B}} \overline{\mathbf{D}}_k (\mathbf{Q}^+)^T, \mathbf{Q} = (\overline{\mathbf{B}}^T \mathbf{B}) \circ (\overline{\mathbf{C}}^T \mathbf{C})$$

$$2 \quad \mathbf{B} = \sum_{k=1}^K \mathbf{X}_k^T \overline{\mathbf{A}} \overline{\mathbf{D}}_k (\mathbf{R}^+)^T, \mathbf{R} = (\overline{\mathbf{A}}^T \mathbf{A}) \circ (\overline{\mathbf{C}}^T \mathbf{C})$$

$$3 \quad \mathbf{C} = \begin{bmatrix} \text{diag}(\overline{\mathbf{A}}^T \mathbf{X}_1 \overline{\mathbf{B}})^T \\ \text{diag}(\overline{\mathbf{A}}^T \mathbf{X}_2 \overline{\mathbf{B}})^T \\ \vdots \\ \text{diag}(\overline{\mathbf{A}}^T \mathbf{X}_K \overline{\mathbf{B}})^T \end{bmatrix} (\mathbf{S}^+)^T, \mathbf{S} = (\overline{\mathbf{A}}^T \mathbf{A}) \circ (\overline{\mathbf{B}}^T \mathbf{B}) \text{ and}$$

where the operator $\text{diag}(\mathbf{M})$ means a column vector containing the diagonal of $\mathbf{M}$, and “ $\circ$ ” is the element-wise Hadamard product.

4 Calculate σ_r

If the relative change in σ_r is small stop.

Otherwise go to step one.

4. APPLICATIONS

4.1 Blind separation of DS-CDMA signals using an antenna array

In [17], it is shown that parallel factor analysis underlies the problem of blind source separation for DS-CDMA systems utilizing receive antenna arrays. Moreover, trilinear modeling provides a unifying framework in which the delicate trade-off between different types of diversity (as required for identifiability) can be understood and thoroughly exploited. In particular, [17] considers blind separation of DS-CDMA signals using an *uncalibrated* antenna array, and subject to

- unknown flat fading;
- unknown multipath;
- quasi-synchronous transmissions;
- over-saturation;
- lack of (or, inexact) knowledge of the spreading codes; and
- loose power control.

A general identifiability result is established, that allows blind recovery of the source signals as well as other system parameters without requiring independence or whiteness, nor finite alphabet or constant modulus. The results apply equally well to either deterministic or stochastic sources; knowledge of the spreading codes is not required; and, actually, certain matrices involved may be allowed to loose rank without sacrificing identifiability. In the noisy case, COMFAC is the algorithm of choice for fitting the complex trilinear model.

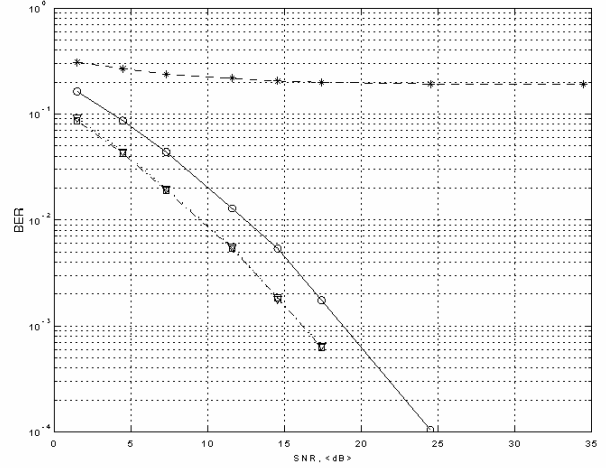


Figure 2.

Figure 2 depicts (Monte-Carlo) average bit error rate versus SNR plots for a DS-CDMA system with four users, two receive antennas, a block length of fifty symbol snapshots, and differentially-encoded BPSK user signals spread using Hadamard(4) codes with two trailing zeros (“guard chips”). Each user undergoes an unknown multipath/delay channel of length three chips. The coefficients of each channel are drawn from an iid Rayleigh generator. We compared against the traditional matched filter receiver (dashed with stars - exploits knowledge of the spreading codes, and averages the independently-faded signal copies received by the two antennas), as well as two *non-blind* approaches: the non-blind minimum mean squared error (MMSE) receiver (dash-dot with squares), and the non-blind zero-forcing (ZF) receiver (double-dot with triangles). Both assume full knowledge of the multipath channels and relative delays, as well as the fading coefficients. In addition, the non-blind MMSE receiver assumes knowledge of SNR. For the proposed method (solid with circles), the factors were estimated using COMFAC. The proposed blind receiver achieves results very close to the non-blind receivers, without knowledge of spreading codes, multipath or SNR.

4.1 Joint Azimuth-Elevation and Signal-Array Response Estimation

In [5], we consider the problem of joint azimuth-elevation, signal, and array response estimation, and show that it too can be cast in the framework of parallel factor analysis. As shown in [5], the results carry over to general setups that include multiple-parameter/multiple-invariance ESPRIT subspace fitting problems.

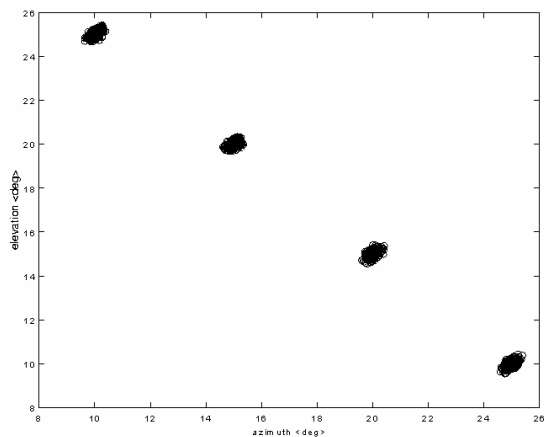


Figure 3.

The application of COMFAC to the joint azimuth-elevation estimation problem is illustrated in Figure 3. It depicts the azimuth-elevation scatter diagram for four sources with (azimuth, elevation) = (10,25), (15,20), (20,15), (25,10) degrees, using COMFAC to fit the trilinear model associated with a 10×10 array with $\lambda/4$ sensor spacing, split into maximal-overlap 81-element sub-arrays, $N = 300$ symbols, and $\text{SNR} = 12$ dB. The statistical performance of COMFAC is an order of magnitude better than sub-optimal approaches to joint azimuth-elevation estimation, and very close to the best methods available utilizing array calibration information.

5. SUMMARY

An effective algorithm has been given for fitting the PARAFAC model to complex-valued data. The algorithm is computationally less costly and more efficient than traditional algorithms, and provides means for blind separation of DS-CDMA signals in unknown multipath [17], joint azimuth-elevation estimation, joint angle-delay estimation, and the subspace fitting formulation of multiple-parameter/multiple-invariance ESPRIT [5].

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